
(12) Practice Problems

Group Lasso. In 2006, Yuan and Lin¹ introduced the group lasso to allow predefined groups of covariates to jointly be selected into or out of a model. This is useful in many settings, perhaps most obviously when a categorical variable is coded as a collection of binary covariates. In this case, group lasso can ensure that all the variables encoding the categorical covariate are included or excluded together. Another setting in which grouping is natural is in biological studies. Since genes and proteins often lie in known pathways, which pathways are related to an outcome may be more significant than whether individual genes are.

(1) Group lasso logistic regression.² Suppose we have features $X \in \mathbb{R}^{n \times (p+1)}$ that we divide into J groups:

$$X = [\mathbf{1}, X_{(1)}, \dots, X_{(J)}],$$

where $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^n$ and each $\dots, X_{(j)} \in \mathbb{R}^{n \times p_j}$. To achieve sparsity over groups of features, rather than individual features, we can use a *group lasso penalty*. Write $\beta = (\beta_0, \beta_{(1)}, \dots, \beta_{(J)}) \in \mathbb{R}^{p+1}$, where β_0 is an intercept term and each $\beta_{(j)} \in \mathbb{R}^{p_j}$. Consider the problem

$$\min_{\beta} g(\beta) + \lambda \sum_{j=1}^J w_j \|\beta_j\|_2,$$

where g is a loss function and $\lambda \geq 0$ is a tuning parameter. The penalty $h(\beta) = \lambda \sum_{j=1}^J w_j \|\beta_j\|_2$ is called the group lasso penalty. A common choice for w_j is $\sqrt{p_j}$ to adjust for the group size.

(a) Derive the proximal operator $\text{prox}_{h,\alpha}(\beta)$ for the group lasso penalty defined above.

(b) Let $y \in \{0, 1\}^n$ be a binary label, and let g be the logistic loss

$$g(\beta) = -\frac{1}{n} \sum_{i=1}^n y_i (X\beta)_i + \frac{1}{n} \sum_{i=1}^n \log(1 + \exp\{(X\beta)_i\}).$$

Write out the steps for proximal gradient descent applied to the logistic group lasso problem.

(c) Now we'll use the logistic group lasso to classify a person's age group from their movie ratings. The movie ratings can be categorized into groups according to a movie's genre (e.g., all ratings for action movies can be grouped together). Load the training data in `trainRatings.txt`, `trainLabels.txt`; the features have already been arranged into groups and you can find information about this in `groupTitles.txt`, `groupLabelsPerRating.txt`. Solve the logistic group lasso problem with regularization parameter $\lambda = 1$ by running proximal gradient descent, Nesterov acceleration and different stochastic methods. Plot $f^{(k)} - f^*$ versus k , where $f^{(k)}$ denotes the objective value at iteration k , and use as an optimal objective value $f^* = 213.741/n$. Make sure the plot is on a semi-log scale (where the y -axis is in log scale).

¹Yuan, Ming; Lin, Yi (2006). "Model Selection and Estimation in Regression with Grouped Variables". Journal of the Royal Statistical Society. Series B (statistical Methodology). 68 (1). Wiley: 49-67.

²Adapted from Ryan Tibshirani's optimization course.

$$g(\beta) = -\frac{1}{n} \sum_{i=1}^n y_i (X\beta)_i + \frac{1}{n} \sum_{i=1}^n \log(1 + \exp\{(X\beta)_i\}).$$

$$\min_{\beta} g(\beta) + \lambda \sum_{j=1}^J w_j \|\beta_j\|_2,$$